Section Id :	64065399594
Section Number :	8
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	2
Number of Questions to be attempted :	2
Section Marks :	25
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653228743
Question Shuffling Allowed :	No

Question Number : 188 Question Id : 6406531416773 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : SPEECH TECHNOLOGY (PEN AND PAPER EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406534740282. ✔ YES

6406534740283. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653228744
Question Shuffling Allowed :	No

Question Number : 189 Question Id : 6406531416774 Question Type : MCQ

Correct Marks : 25

Question Label : Multiple Choice Question

1. (1 point) A speech recognition task has **100 labeled hours** and **1000 unlabeled hours** of audio. Which approach is most likely to achieve the **best generalization** in the downstream tasks?

   A) Train only on the 100 labeled hours using a supervised acoustic model

   B) Perform k-means clustering on the 1000 unlabeled hours and use cluster IDs as labels

   C) Pretrain a model on 1000 hours with self-supervised objectives, then fine-tune on the 100 labeled hours

   D) Train a text-only language model on transcriptions of the 100 hours

2. (1 point) Imagine that you design a closed-loop system where speech is synthesized from text (TTS) and then transcribed back into text (ASR). Which evaluation measure best captures the accuracy of the text reconstruction at the end of this loop?

   A) Signal-to-noise ratio (SNR) of the reconstructed audio

   B) Word Error Rate (WER) between the original and reconstructed text

   C) Cosine similarity between mel-spectrograms of original and reconstructed audio

   D) Perplexity of the reconstructed text under a language model

3. (1 point) For an image captioning system, which combination of encoder and decoder is most suitable?

   A) Encoder: RNN, Decoder: CNN

   B) Encoder: CNN, Decoder: RNN (e.g., LSTM/GRU)

   C) Encoder: Transformer, Decoder: CNN

   D) Encoder: MLP, Decoder: RNN

4. (1 point) Which system is most likely to fail if the same sentence is spoken by two different speakers with identical accents?

   A) ASR

   B) Speaker Verification

   C) TTS

   D) MT

5. (1 point) Which of the following is shared across different positions in a Transformer's self-attention mechanism?

   A) Queries

   B) Keys

   C) Values

   D) Weight matrices

6. (1 point) Which mechanism in Transformers is responsible for transferring source sequence information into target generation?

   A) Positional Encoding

   B) Encoder-Decoder Attention

   C) Feed Forward Layers

   D) Layer Normalization

7. (1 point) In seq2seq with RNNs, attention was introduced mainly because:

   A) RNNs could not handle character-level input

   B) Fixed-length context vectors limited long sequence handling

   C) RNNs always produced the wrong output sequence length

   D) Attention made RNNs faster than Transformers

8. (2 point) You trained Word2Vec on a legal corpus, but the new word *"cybercrime"* appears rarely. Consider the following statements:

   (a) Word2Vec will directly generate a reliable embedding for *"cybercrime"*.

   (b) Subword-based models (e.g., FastText) can form embeddings from character n-grams like *"cyber"* and *"crime"*.

   (c) Contextual models such as BERT can still assign meaningful embeddings to *"cybercrime"* by leveraging surrounding context and subword tokenization.

   Which of the above statements are correct?
   1) Only (b)
   2) Only (c)
   3) Both (b) and (c)
   4) All of the above

9. (3 points) CNN Kernels

   You have a 7×7 input image and apply two successive convolutions with 3×3 kernels, stride $= 2$, padding $= 1$. What is the final output size?

10. (3 points) BERT applies the Masked Language Modeling (MLM) strategy by randomly masking 15% of tokens in a sequence. Consider a sentence of length 20 tokens. Answer the following:

    A) Calculate the expected number of tokens selected for masking. (1 point)

    B) Among these masked positions: 80% are replaced by `[MASK]`, 10% by a random token, and 10% are left unchanged. Derive the expected counts for each case (round suitably). (1 point)

    C) Why does BERT deliberately keep some tokens unchanged during MLM training, instead of masking all? (1 point)

11. (5 points) You are designing a CNN for medical imaging. Compare the effects of increasing the kernel size from 3×3 to 7×7.

    (a) Derive an effective receptive field after 3 layers of 3×3 vs 1 layer of 7×7. (2 marks)

    (b) Which captures more local context vs global context? (1 mark)

    (c) Why might stacking smaller kernels be preferred despite same receptive field? (2 marks)

12. (5 points) Transformers

(a) (2 points) Explain the self-attention mechanism in Transformers. Specifically describe the role of the Query (Q), Key (K), and Value (V) matrices in computing attention weights.

(b) (3 points) Given:

$$Q = [2, 1], \quad K = \begin{bmatrix} 1 & 1 \\ 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

(a) Compute raw attention scores QK^T. (1 point)

(b) Apply softmax. (1 point)

(c) Compute output = weights $\cdot V$ and Interpret which key the query attends most. (1 point)

Options :

6406534740284. ✔ I have written answers on the answer sheets

6406534740285. ✖ Not applicable

SPG

Section Id :	64065399595
Section Number :	9
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	18
Number of Questions to be attempted :	18
Section Marks :	30
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653228745
Question Shuffling Allowed :	No

Question Number : 190 Question Id : 6406531416775 Question Type : MCQ

Correct Marks : 0